

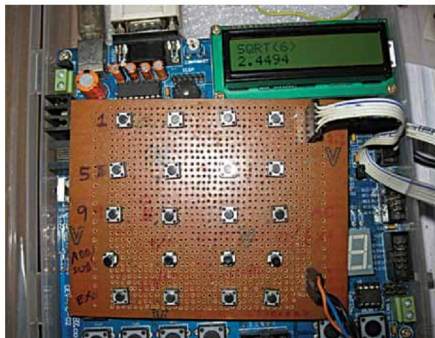


Fig. 10: Working prototype of scientific calculator

help of timer0. When '+/-' switch is pressed once, the LCD shows '+'. When it is pressed again within a short period of time, the LCD shows '-'.

When it is pressed again within a short period of time, the LCD shows '-'.

Pressing it further shows '+'. After reset, '50000' is shown on the LCD. Initially, the cursor is at 'S'. At this time, no switch other than '+/-' works. This '+/-' switch is time-multiplexed, and either '+' or '-' can be displayed at the same location of the LCD.

As you input '+' or '-' from the keypad, the cursor moves to the next position. The four zeros (0000) allow you to input the 4-digit number from the keypad. The digits are taken one by one. Then '.00' is displayed on the LCD, allowing you to input digits in decimal places as well. Pressing the 'AC' key while executing the code clears the LCD and the program starts executing from the beginning.

Fig. 9 shows the user manual of the calculator. Fig. 10 shows the author's working prototype.

EFY note. The source code of this article is available on www.efymag.com website. ●

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